

Engineering Change Order ECO-2026-008

Product: ACT-ST-10 Wireline Pressure Setting Tool | Revision A → B

ECO Number / Date	ECO-2026-008 / 2026-03-02
Originator	A. Gutierrez
Affected Product	ACT-ST-10 Wireline Pressure Setting Tool
Revision Change	A to B
Change Classification	Class 2 (material substitution, form/fit identical)
Effectivity	All redresses performed after 2026-03-02.

1. Reason for Change

Winter operations in the Bakken (ambient to -15 degF) produced two misruns traced to firing-head o-ring leakage. Viton (FKM) -116 o-rings stiffen below +5 degF; gas bypass at the initiator adapter prevented full stroke.

2. Description of Change

Change all ST-10 dynamic and firing-head o-rings from Viton 75/90A to HNBR 90A (ACT-ORG-116-H90, -210-H90, -218-H90). HNBR compound ACT-H90N rated -25 degF to 300 degF. Viton remains available as a high-temperature option kit (ACT-ST-10-RK-HT) for wells above 300 degF. Redress kit ACT-ST-10-RK content updated.

3. Affected Documents

Document	From Rev	To Rev	Change Summary
BOM-ST-10	A	B	O-ring part numbers
MAN-ST-10	A	B	Sec 4 redress note
DS-ST-SERIES	A	B	Materials table
Redress kit ACT-ST-10-RK	-	-	Content change

4. Validation / Qualification Evidence

Low-temperature function test at -25 degF soak: 5 firings, full stroke, no leakage (Test report TR-26-018). Compound ACT-H90N certified per batch (see elastomer certificate EC-HNBR-B25-0339 format).

5. Disposition of Existing Stock and WIP

Viton kits in stock: relabel as -HT option. Tools in field: convert at next redress.

6. Implementation Checklist

Task	Owner	Status
Update drawings and BOM in PLM	Design Engineering	COMPLETE
Update datasheet and manual	Document Control	COMPLETE
Update router / work instructions	Manufacturing Engineering	COMPLETE
Train assembly & QC personnel	Quality	COMPLETE
Notify affected customers / bulletin	Sales Engineering	COMPLETE

Task	Owner	Status
First-article inspection on next lot	Quality	SCHEDULED

7. Approvals

Role	Name	Signature	Date
Engineering	J. Martinez, P.E.	/signed electronically/	2026-02-27
Quality	S. Nguyen	/signed electronically/	2026-03-01
Manufacturing	T. Okafor	/signed electronically/	2026-03-02
Doc Control	L. Hargrove	/signed electronically/	2026-03-02